

Generalized Robinson Correspondence via Fomin's Growth Diagrams

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Introduction

The classical Robinson correspondence asserts a bijection between the symmetric group S_n and pairs of standard Young tableaux of the same shape $\lambda \vdash n$. This celebrated correspondence, together with its generalization—the Robinson–Schensted–Knuth correspondence—plays a fundamental role in combinatorics, representation theory, and symmetric function theory. Remarkably, these bijections are not restricted to the Young lattice, but extend to a wider class of posets called *differential posets*, introduced by Richard Stanley. Independently, Fomin studied this class under the name of *Y-graphs* and developed a framework to generalize the Robinson correspondence using local growth rules. As a consequence, many classical results arise naturally in this setting.

In these notes, we present Fomin’s approach to constructing such bijections for differential posets. We mainly focus on two fundamental examples: the Young lattice and the Young Fibonacci lattice. In the case of the Young lattice, one has several equivalent constructions of the bijection, including the classical insertion algorithm, the matrix-ball construction (Viennot’s light-and-shadow method), and a global rule on vertices arising from Greene’s theorem 5.1. In addition, there are local rules on vertices and edges defined via growth diagrams. All these constructions yield the same bijection, each offering its own advantages.

The Young Fibonacci lattice provides another important example, where analogous global and local rules, as well as insertion-type algorithms, can be developed, again leading to the same bijection.

The material presented here is standard and is largely based on [1], [3], [2], and [4].

1 The Young Lattice and the Young–Fibonacci Lattice

We begin with the Young lattice and the Young–Fibonacci lattice, which are the main examples considered in these notes. We recall their definitions and use them to motivate the general form of Robinson correspondence.

Definition 1.1. Let G be an oriented graph with vertex set $V(G)$. A map $|\cdot|: V(G) \rightarrow \mathbb{Z}$ is a *grading* if for every edge $g_1 \rightarrow g_2$ we have $|g_2| = |g_1| + 1$. We call G a *graded graph*.

1.1 The Young Lattice

Let $\mathbb{Y}$ denote the *Young lattice*: the poset of integer partitions λ , where $\mu \leq \lambda$ if and only if λ/μ is a single box. The Young lattice is a graded poset with the grading the number of boxes.

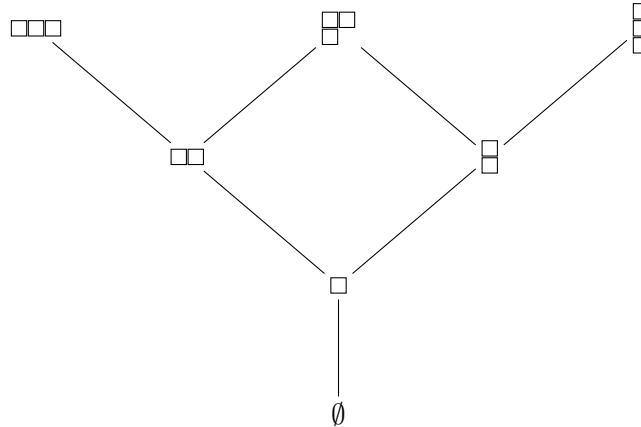


Figure 1: The Young’s lattice $\mathbb{Y}$ (levels 0–3).

1.2 The Young–Fibonacci Lattice

Let V be the set of finite words in the alphabet $\{1, 2\}$. The *Young–Fibonacci lattice* $\mathbb{Y}F$ is the poset with covering relation: $s < w$ if s is obtained from w by removing the first letter 1 or replacing any 2s before the first 1 by 1. Equivalently we get the elements at the n th level (sum of letters is n) recursively as follows; it is the union of elements obtained by prepending 2 to the elements at the $(n-2)$ th level or prepending 1 to the elements at the $(n-1)$ th level. Thus the number of elements at the n th level is the n th Fibonacci number. The Young–Fibonacci lattice is also a graded graph with the grading the sum of letters of each word.

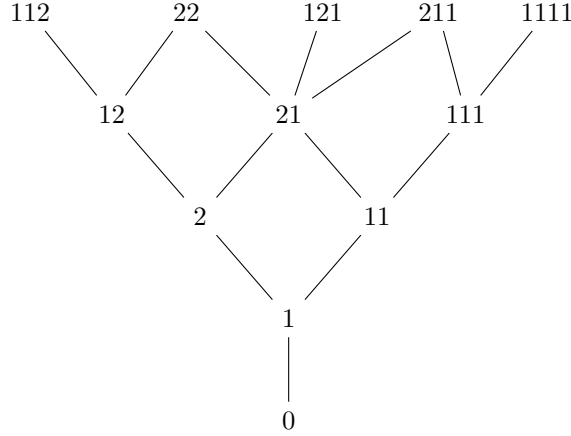


Figure 2: The Young–Fibonacci lattice $\mathbb{Y}F$ (levels 0–4).

1.3 Motivation for Robinson Correspondence in $\mathbb{Y}$ and $\mathbb{Y}F$

For a vertex λ in a graded poset P , let $f_\lambda = \#\{\text{paths from } 0 \text{ to } \lambda\}$.

Example 1.2 (Young lattice). At level 3, we have

$$f_{(3)} = 1, \quad f_{(2,1)} = 2, \quad f_{(1,1,1)} = 1$$

and

$$\sum_{|\lambda|=3} f_\lambda^2 = 1^2 + 2^2 + 1^2 = 6 = 3!.$$

Similarly, at level 4, we have

$$f_{(4)} = 1, \quad f_{(3,1)} = 3, \quad f_{(2,2)} = 2, \quad f_{(2,1,1)} = 3, \quad f_{(1,1,1,1)} = 1$$

and

$$\sum_{|\lambda|=4} f_\lambda^2 = 1^2 + 3^2 + 2^2 + 3^2 + 1^2 = 24 = 4!.$$

Example 1.3 (Young–Fibonacci lattice). At level 3,

$$f_{12} = 1, \quad f_{21} = 2, \quad f_{111} = 1.$$

Thus

$$\sum_{|\lambda|=3} f_\lambda^2 = 1^2 + 2^2 + 1^2 = 6 = 3!.$$

At level 4,

$$f_{1111} = 1, \quad f_{211} = 3, \quad f_{121} = 2, \quad f_{112} = 1, \quad f_{22} = 2.$$

Thus

$$\sum_{|\lambda|=4} f_\lambda^2 = 1^2 + 3^2 + 2^2 + 1^2 + 2^2 = 24 = 4!.$$

These examples suggest a pattern which holds more generally. For the Young lattice, this could be explained by the classical Robinson correspondence: permutations in S_n are in bijection with pairs of paths starting at $\emptyset$ and ending at the same vertex at level n . This could be seen in the following manner as well: because the bases for the irreducible representations of S_n are indexed by standard Young tableaux, A lattice path from $\emptyset$ to λ corresponds to a standard tableau of shape $\lambda \vdash n$. It is classical that

$$\sum_{\lambda \vdash n} d_\lambda^2 = n!, \quad (1)$$

where d_λ is the dimension of the irreducible S_n -representation indexed by λ .

However, we shall see that Fomin's growth diagrams give a general way to construct such bijections for a wide class of posets, called *differential posets*, which include both $\mathbb{Y}$ and $\mathbb{Y}F$.

2 Growth Diagrams and Differential Posets

In this section we introduce the language needed to state Fomin's construction. We first describe growth diagrams and then recall differential posets, the class of posets where the local rules will be applied.

2.1 Skew Diagrams and Growths

Definition 2.1. Consider $\mathbb{Z}^2$ with grading $|(x, y)| = x + y$ and partial order $(x_1, y_1) \leq (x_2, y_2)$ if $x_1 \leq x_2$ and $y_1 \leq y_2$. A finite convex subset $S \subseteq \mathbb{Z}^2$ (i.e., $a, c \in S$ and $a \leq b \leq c \Rightarrow b \in S$) is called a *skew diagram*. The *upper/lower boundary* of S :

$$\partial^+(S) = \{(x, y) \in S : (x+1, y+1) \notin S\}, \quad \partial^-(S) = \{(x, y) \in S : (x-1, y-1) \notin S\}.$$

One important example of a skew diagram is

$$Sq_n = \{(x, y) \in \mathbb{Z}^2 : 0 \leq x \leq n, 0 \leq y \leq n\}.$$

Definition 2.2. A *generalized permutation* σ is a finite set of cells of S such that no two lie in the same row or column.

Definition 2.3. Let P and Q be two graded posets. A map $g : P \rightarrow Q$ is called a *growth* if:

$$x \leq y \implies g(x) \leq g(y).$$

Example 2.4. 1. The rank function $\rho : P \rightarrow \mathbb{Z}$.

2. A multichain in P is a growth from $\mathbb{Z}$ to P .

3. The composition of two growths.

Definition 2.5. A growth on $\mathbb{Z}^2$ is called a *2-dimensional growth*.

Example 2.6. Below is a map $g : S \rightarrow \mathbb{Z}$, where $g(v) = \#$ cells marked with an X which are below and to the left of vertex v . This is a two dimensional growth.

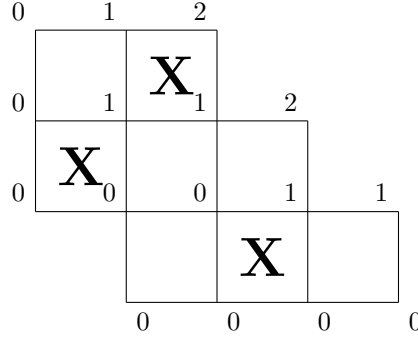


Figure 3

But the following is not a growth.

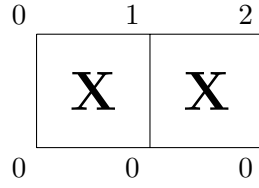


Figure 4

2.2 Differential Posets

Definition 2.7. A graded poset P is a *differential poset* if:

- (1) P is locally finite (i.e., for any $x, y \in P$, $[x, y]$ is finite), graded, with $\hat{0}$.
- (2) If $(x, y) \in P \times P$ with $x \neq y$, then the sets $C^+(x) \cap C^+(y)$ and $C^-(x) \cap C^-(y)$ are either both empty or each consists of one vertex where

$$C^+(x) = \{y \in P : x \leq y\} \quad \text{and} \quad C^-(x) = \{y \in P : y \leq x\}.$$

(modularity condition).

- (3) $|C^+(x)| = |C^-(x)| + 1$ for every $x \in P$.

Lemma 2.8. $\mathbb{Y}$ and $\mathbb{Y}F$ are differential posets.

Proof. We first prove the result for the Young lattice $\mathbb{Y}$. It is locally finite, graded by the number of boxes, and has the unique minimal element $\emptyset$.

We check condition (2). Let λ and μ be two distinct partitions. Suppose that λ and μ have a common upper cover ν . Then λ and μ are obtained from ν by deleting one box each. Since $\lambda \neq \mu$, these deleted boxes are distinct. Hence λ and μ differ by moving one box from one row to another. In this case the common upper cover is forced: it is obtained by adding both of these boxes, or equivalently it is $\lambda \cup \mu$. Thus $C^+(\lambda) \cap C^+(\mu)$ has exactly one element. Similarly, the common lower cover is forced: it is obtained by deleting both of these boxes, or equivalently it is $\lambda \cap \mu$. Hence $C^-(\lambda) \cap C^-(\mu)$ also has exactly one element. The same argument starting with a common lower cover gives the converse direction. Therefore the two sets $C^+(\lambda) \cap C^+(\mu)$ and $C^-(\lambda) \cap C^-(\mu)$ are either both empty or both consist of one element.

We now check condition (3) for the Young lattice. Let λ be a partition, and suppose that λ has d distinct nonzero row lengths. The elements covered by λ are obtained by deleting one removable box from λ . Such a removable box occurs exactly at the end of a row whose length

is strictly larger than the row below it. Hence there are exactly d elements covered by λ , i.e. $|C^-(\lambda)| = d$. On the other hand, the elements covering λ are obtained by adding one box to λ . This can be done at the end of any row whose length is strictly smaller than the row above it. In addition, we may add a box to a new row below the diagram. Thus there are $d + 1$ possible additions. Hence $|C^+(\lambda)| = d + 1$. Thus $\mathbb{Y}$ is a differential poset.

Now consider the Young–Fibonacci lattice $\mathbb{Y}F$. It is locally finite, graded by the sum of the letters of the word, and has the unique minimal element 0.

By construction, for any word $x \in \mathbb{Y}F$,

$$C^+(x) = \{1x\} \cup \{2z : z \in C^-(x)\}.$$

Indeed, an element covering x is obtained either by adding a 1 at the beginning, or by putting a 2 in front of an element covered by x . This immediately gives $|C^+(x)| = |C^-(x)| + 1$, so condition (3) holds.

It remains to check condition (2). Let $x \neq y$ be two elements of $\mathbb{Y}F$. If they have different ranks, then they cannot have a common upper cover or a common lower cover. So assume that they have the same rank. Using the description above, we get

$$C^+(x) \cap C^+(y) = \{2z : z \in C^-(x) \cap C^-(y)\}.$$

Hence the number of common upper covers of x and y is the same as the number of common lower covers.

We now show that this number is at most one. This follows by induction on the rank. The assertion is clear in the lowest ranks. Suppose it holds in all lower ranks. If x and y had two distinct common lower covers z_1 and z_2 , then x and y would be two distinct common upper covers of z_1 and z_2 , contradicting the induction hypothesis. Thus $C^-(x) \cap C^-(y)$ has at most one element, and hence so does $C^+(x) \cap C^+(y)$. Thus $\mathbb{Y}F$ is a differential poset. $\square$

2.3 Semimodular Growths

In this section we would like to characterize those two dimensional growths which come from a generalized permutation.

Definition 2.9. A 2-dimensional growth $g : S \rightarrow \mathbb{Z}$ is called *semimodular* if the values of g on the vertices of any cell, say $g_{00}, g_{01}, g_{10}, g_{11}$ (see the figure 5) satisfy the following inequality:

$$g_{00} + g_{11} \geq g_{01} + g_{10}.$$

A cell for which the above inequality is strict is called an *atom* of g .

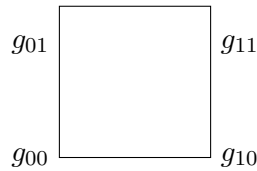


Figure 5

Example 2.6 shows a semimodular growth with atoms marked by X. We follow the convention of marking the atoms by an X for a semimodular growth. It is not hard to see that the set of atoms of a semimodular growth on S is a generalized permutation in S .

Lemma 2.10. Any semimodular growth $g : S \rightarrow \mathbb{Z}$ is uniquely determined by its restriction to its lower boundary $\partial^-(S)$ together with a generalized permutation, which is the set of atoms of g .

Proof. We sketch the proof. Locally, there are three possible cases.

- $g_{00} = g_{11} =: a$. Then by the definition of a semimodular growth, $g_{01} = g_{10} = a$.
- $g_{11} = g_{00} + 2 =: a + 2$. Then $g_{01} = g_{10} = a + 1$.
- $g_{11} = g_{00} + 1 =: a + 1$. This has three possibilities. If the cell is marked with an X, then $g_{01} = g_{10} = a$. If the cell is not marked with an X, then either $g_{01} = a, g_{10} = a + 1$ or $g_{01} = a + 1, g_{10} = a$.

Thus the values g_{00}, g_{01}, g_{10} , together with a generalized permutation, determine g_{11} uniquely. $\square$

In the lemma 2.10, we can replace the lower boundary by upper boundary and the proof goes similarly.

3 Fomin's Local Growth Rules

In this section we describe Fomin's growth rules, which are the main ingredient in constructing the generalized Robinson correspondence. We then explain how these rules lead to the correspondence and record a few consequences.

3.1 R-Correspondences and Fomin's Theorem

In this section we try to extend a growth on a skew diagram S uniquely from its lower boundary or upper boundary to the entire S like we did for semimodular growth. We might find some difficulties while doing that. To bypass these difficulties we shall introduce the notion of an R -correspondence.

Let $g : S \rightarrow P$ be a growth, where P is a differential poset. We look at one cell of S , with vertex values denoted as in Figure 5. Suppose first that we know g_{01}, g_{10} , and g_{11} , and want to determine g_{00} . There are a few cases.

Case 1. If $|g_{01}| < |g_{10}|$, then the growth condition forces $g_{00} = g_{01}$, $g_{11} = g_{10}$.

Case 2. If $|g_{10}| < |g_{01}|$, similarly we must have $g_{00} = g_{10}$, $g_{11} = g_{01}$.

Case 3. Suppose $|g_{01}| = |g_{10}|$ and $g_{01} \neq g_{10}$. Then by the modularity condition (2) in the definition of a differential poset, the common lower cover is uniquely determined. Hence $g_{00} = g_{01} \cap g_{10}$.

Case 4. The only remaining case is $g_{01} = g_{10} = x$. This is the case where an extra choice is needed. Since P is differential, we have $|C^+(x)| = |C^-(x)| + 1$. Thus $C^+(x)$ has one more element than $C^-(x)$. To deal with this, we choose a bijection $\Phi_x : C^+(x) \rightarrow C^-(x) \cup \{x\}$. We also use the convention $\Phi_x(x) = x$. Then, when extending the growth downwards, we define $g_{00} = \Phi_x(g_{11})$. This gives the required value of g_{00} . It is in fact not hard to see that this growth is semimodular.

Now consider the reverse problem, where we want to extend the growth upwards. The first three cases are again forced. In the fourth case, if $g_{00} \neq x$, then $\Phi_x^{-1}(g_{00})$ determines g_{11} . The ambiguity occurs when $g_{00} = g_{01} = g_{10} = x$. In this situation, we use the generalized permutation: if the cell is not marked, we set $g_{11} = x$; if the cell is marked by an X, we choose the element $y \in C^+(x)$ such that $\Phi_x(y) = x$, and set $g_{11} = y$. This is the local rule which allows the growth to be extended cell by cell. Once again, the growth we obtain so is semimodular.

Definition 3.1. An R -correspondence on a differential poset P is a collection $\Phi = \{\Phi_x\}_{x \in P}$ of bijections $\Phi_x : C^+(x) \rightarrow C^-(x) \cup \{x\}$. A growth $g : S \rightarrow P$ is a Φ -growth if whenever $g_{01} = g_{10}$, we have $g_{00} = \Phi_{g_{01}}(g_{11})$.

From what we have discussed above, we obtain the following lemmas and theorem.

Lemma 3.2. *Let P be a differential poset and S be a skew diagram. Any growth $g^+ : \partial^+(S) \rightarrow P$ can be uniquely extended to a Φ growth g on all of S . Furthermore $|g|$ will be semimodular.*

Lemma 3.3. *Let P be a differential poset, S a skew diagram, and let σ be a generalized permutation on the cells of S . Let $f^- : \partial^-(S) \rightarrow P$ be a growth on the lower boundary of S . Then there is a unique Φ -growth $g : S \rightarrow P$ extending f^- such that the atoms of $|g|$ are exactly the cells of σ and $|g|$ is semimodular.*

Theorem 3.4 (Fomin [3]). *Let P be a differential poset, S a skew diagram, Φ an R -correspondence. Then we have a bijection*

$$\{\text{growths } g^+ : \partial^+(S) \rightarrow P\} \longleftrightarrow \{(g^- : \partial^-(S) \rightarrow P, \sigma)\},$$

where σ is a generalized permutation of S . Each such pair is uniquely associated with a Φ -growth $g : S \rightarrow P$, whose restrictions to the upper and lower boundaries are g^+ and g^- , respectively. Moreover, $|g|$ is semimodular and its atoms are precisely the cells of σ .

Note that the extension is unique with respect to a fixed Φ growth.

3.2 Scheme of Generalized Robinson Correspondence

We now explain how Fomin's theorem gives the generalized Robinson correspondence. Let P be a differential poset with a fixed R -correspondence Φ , and let S be a skew diagram. Given a growth on the upper boundary $g^+ : \partial^+(S) \rightarrow P$, Fomin's local rules extend it uniquely to a Φ -growth $g : S \rightarrow P$. The modulus $|g|$ is semimodular, and hence its atoms form a generalized permutation on the cells of S . Thus a growth on the upper boundary naturally determines a generalized permutation.

The fundamental special case is $S = Sq_n$. Then for a differential poset P , each pair of paths of length n , going from 0 to the same vertex of the n th level of the graph P can be interpreted as a growth $\partial^+(Sq_n) \rightarrow P$. Under this interpretation, the upper boundary growth is strict if and only if the corresponding generalized permutation is an ordinary permutation in S_n . Thus by theorem 3.4, fixing the growth on lower boundary to 0, we obtain a bijection between S_n and a pair of paths in P starting at 0 and ending at the same vertex at the n -th level. Hence the classical Robinson correspondence.

3.3 Applications of Robinson Correspondence

We would like to introduce an immediate application of Fomin's theorem 3.4. Let the skew diagram S have the following form (figure 6):

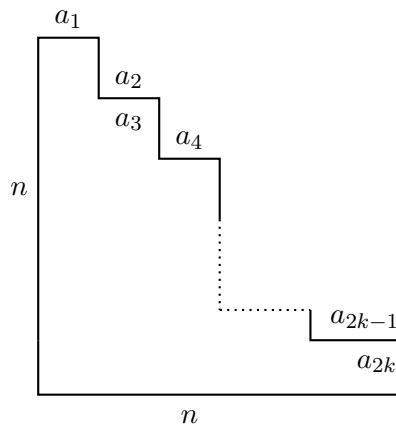


Figure 6

We want to count closed walks in P which start and end at 0, and whose steps have the prescribed form: first a_1 upward steps, then a_2 downward steps, then a_3 upward steps, and so on. Such a walk is the same as a growth $g^+ : \partial^+(S) \rightarrow P$ on the upper boundary of S . If we fix the lower boundary growth to be $g^- = 0$, then Fomin's theorem gives a bijection between these walks and generalized permutations on the cells of S .

We also record some standard consequences of the correspondence in the special case $S = Sq_n$. For a differential poset P , Fomin's theorem gives a bijection between permutations in S_n and pairs of paths starting at 0 and ending at the same vertex at level n .

Corollary 3.5. *If we have the following bijection by Robinson correspondence;*

$$\pi \xleftrightarrow{R} (P, Q) \quad \text{then} \quad \pi^{-1} \xleftrightarrow{R} (Q, P)$$

Proof. Replacing π with π^{-1} corresponds to interchanging the rows and columns of our skew diagram (Sq_n in our case), i.e., transposing the diagram. In this case it is clear that the entire correspondence is transposed, switching P and Q . $\square$

If π is an involution, then $\pi = \pi^{-1}$. By the previous corollary, its corresponding pair must be of the form (P, P) . Therefore, we have the following corollary.

Corollary 3.6. *The number of involutions in S_n is $\sum_{|\lambda|=n} f_\lambda$ where f_λ denotes the number of paths in the differential poset P from 0 to λ .*

By the Robinson correspondence, permutations in S_n are in bijection with pairs of paths (P, Q) starting at 0 and ending at the same vertex λ of level n ; hence counting both sides gives:

Corollary 3.7. *For every $n \in \mathbb{N}$, $\sum_{|\lambda|=n} f_\lambda^2 = n!$.*

4 Fomin's Local Growth Rules on the Young Lattice

In this section we apply the theory developed so far to Young lattice. Here we get the Robinson correspondence as a consequence of Fomin's theorem 3.4. For this purpose, we need to fix a Φ correspondence. Thus we get the bijection by local rule on vertices.

We first choose an R -correspondence for the Young lattice $\mathbb{Y}$. Let $\mu < \lambda$, and suppose that λ is obtained from μ by adding a box in the i -th row. We define $\Phi_\mu(\lambda) = \nu$ as follows. If $i = 1$, then $\nu = \mu$. If $i > 1$, then ν is obtained from μ by deleting the last box of the $(i - 1)$ -st row. Equivalently, in terms of Young diagrams, Φ_μ records the following operation: when the added box is in row i , we delete a box from the row immediately above it, whenever such a row exists. If the added box is in the first row, we simply get μ .

With reference to the above Φ growth, the local rules on vertices can be easily written in the following manner:

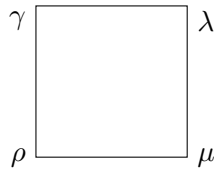


Figure 7

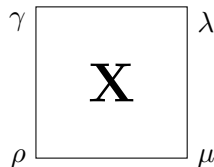


Figure 8

- (1) $\rho = \mu = \gamma \Rightarrow \lambda = \gamma$ (in case of figure 7).
- (2) $\rho = \mu \neq \gamma \Rightarrow \lambda = \mu$ (figure 7).
- (3) $\rho = \gamma \neq \mu \Rightarrow \lambda = \mu$ (figure 7).
- (4) ρ, μ, γ pairwise not equal $\Rightarrow \lambda = \mu \cup \gamma$
- (5) $\rho \neq \mu = \gamma$: if $\rho_j = \mu_j$ except i with $\rho_i = \mu_i - 1$, then $\lambda_j = \mu_j$ except $\lambda_{i+1} = \mu_{i+1} + 1$ (figure 7).
- (6) $\rho = \mu = \lambda$ and X not in box: $\rho = \mu = \gamma = \lambda$ (figure 7).
- (7) $\rho = \mu = \gamma$, and X present: $\lambda = \mu + 1$, i.e., a box is added to first row of μ (figure 8).

Example 4.1. Consider the permutation

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 7 & 1 & 5 & 6 & 4 & 3 \end{pmatrix}.$$

Then fixing $\emptyset$ on the lower boundary of Sq_7 and applying Fomin's local growth rules above, we obtain the following growth diagram:

$\emptyset$	1	2	21	22	32	321	3211
		X					
$\emptyset$	1	1	11	21	31	32	321
					X		
$\emptyset$	1	1	11	21	21	22	221
				X			
$\emptyset$	1	1	11	11	11	21	22
						X	
$\emptyset$	1	1	11	11	11	11	21
							X
$\emptyset$	1	1	11	11	11	11	11
	X						
$\emptyset$	$\emptyset$	$\emptyset$	1	1	1	1	1
			X				
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

Figure 9: Growth diagram for the permutation π in the Young lattice

Since paths in the Young lattice starting at $\emptyset$ are equivalent to standard Young tableaux, the boundary paths give the following pair of tableaux corresponding to π .

$$P = \begin{array}{|c|c|c|} \hline 1 & 3 & 6 \\ \hline 2 & 4 & \\ \hline 5 & & \\ \hline 7 & & \\ \hline \end{array}, \quad Q = \begin{array}{|c|c|c|} \hline 1 & 2 & 5 \\ \hline 3 & 4 & \\ \hline 6 & & \\ \hline 7 & & \\ \hline \end{array}$$

5 Equivalent Constructions for the Young Lattice

In this section we shall see at least four ways other than Fomin's growth rule on vertices to obtain the classical Robinson bijection between the set of pair of standard tableaux of same shape $\lambda \vdash n$ and permutations in S_n .

5.1 Row Bumping

One can read about the general algorithm for row bumping from section 1.1 of [2]. Given

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 7 & 1 & 5 & 6 & 4 & 3 \end{pmatrix},$$

insert values using the row-bumping rule:

$$\emptyset \xrightarrow{2} \begin{array}{|c|} \hline 2 \\ \hline \end{array} \xrightarrow{7} \begin{array}{|c|c|} \hline 2 & 7 \\ \hline \end{array} \xrightarrow{1} \begin{array}{|c|c|} \hline 1 & 7 \\ \hline 2 & \\ \hline \end{array} \xrightarrow{5} \begin{array}{|c|c|} \hline 1 & 5 \\ \hline 2 & 7 \\ \hline \end{array} \xrightarrow{6} \begin{array}{|c|c|c|} \hline 1 & 5 & 6 \\ \hline 2 & 7 & \\ \hline \end{array} \xrightarrow{4} \begin{array}{|c|c|c|} \hline 1 & 4 & 6 \\ \hline 2 & 5 & \\ \hline 7 & & \\ \hline \end{array} \xrightarrow{3} P = \begin{array}{|c|c|c|} \hline 1 & 3 & 6 \\ \hline 2 & 4 & \\ \hline 5 & & \\ \hline 7 & & \\ \hline \end{array}$$

Recording tableau: $Q = \begin{array}{|c|c|c|} \hline 1 & 2 & 5 \\ \hline 3 & 4 & \\ \hline 6 & & \\ \hline 7 & & \\ \hline \end{array}$

Given a pair of standard tableaux, we have the reverse bumping process to obtain a permutation corresponding to the pair which is also discussed in [2]. This maps are inverses of each other, thus giving a required bijection.

5.2 Matrix Ball Construction

Take the same $\pi \in S_7$ as before. Place balls on the permutation matrix A at positions $(i, \pi(i))$. For π above:

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Now use the matrix ball construction to get the pair (P, Q) . This is based on northwest numbering rule and doing it repeatedly. The process of matrix ball construction is schematically shown as:

Here the i th entry of the first row of P is the number of the left most column where a circle with number i occurs, i th entry of the second row of P is the left most column where a square labeled i occurs. Similarly the later rows of P is read from the diagram in the order of triangles and diamond respectively. Do the same process with rows for reading the rows of Q .

This process with its inverse is discussed in chapter 4 of [2]. Interestingly, we get the same bijection as in section 2.1.

$$P = \begin{array}{|c|c|c|} \hline 1 & 3 & 6 \\ \hline 2 & 4 & \\ \hline 5 & & \\ \hline 7 & & \\ \hline \end{array}, Q = \begin{array}{|c|c|c|} \hline 1 & 2 & 5 \\ \hline 3 & 4 & \\ \hline 6 & & \\ \hline 7 & & \\ \hline \end{array}$$

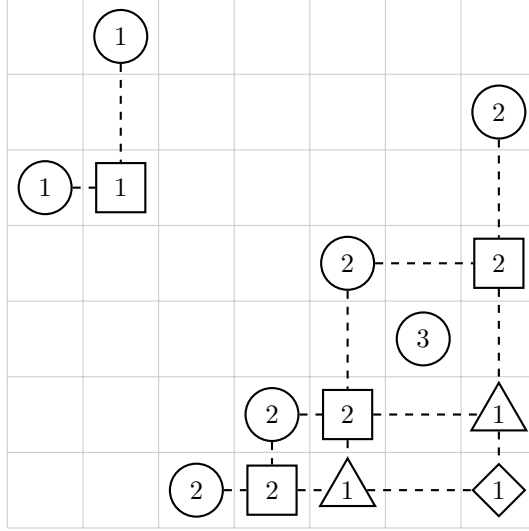


Figure 10: Matrix ball construction

5.3 Global Rule for Young Lattice

In this subsection we associate a permutation to a pair of standard tableaux by means of assigning Young diagrams to each vertex of the skew diagram corresponding to the given permutation. This is achieved with the help of the following theorem.

Theorem 5.1 (Greene, [3]). *Let P be a finite poset. Set $\lambda_k(P) = c_k(P) - c_{k-1}(P)$ and $\mu_k(P) = a_k(P) - a_{k-1}(P)$, where $c_k(P)$ (resp. $a_k(P)$) is the size of the longest union of k chains (resp. antichains). Then $\lambda(P)$ and $\mu(P)$ are partitions, and μ is the conjugate of λ .*

Example 5.2. Consider the following poset.

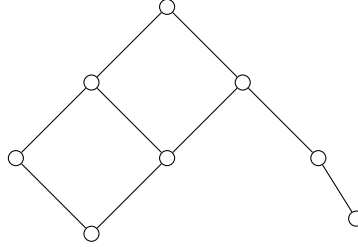


Figure 11

Corresponding to the theorem 5.1, the values of $c_k(P)$ are $c_0 = 0$, $c_1 = 4$, $c_2 = 7$, $c_3 = 8$, $c_4 = 8, \dots$ thus the Young diagram corresponding to this poset has shape $(4, 3, 1)$.

Now it is the time to see the global rule for finding a pair of standard tableaux corresponding to a permutation. For any vertex v of S , define $C(v)$ to be the set of cells of S below and left to v . Also note that the set of cells of S form a poset in an obvious manner (a cell is less than other if it lies weakly left and weakly below to the latter). Let σ be a generalized permutation and for a poset P , $\lambda(P)$ denote the Young diagram corresponding to the poset obtained using Greene's theorem 5.1.

Define a growth $G : Sq_7 \rightarrow \mathbb{Y}$ (where $\mathbb{Y}$ is the Young lattice)

$$v \mapsto \lambda(C(v) \cap \sigma).$$

This is a 2-dimensional growth.

Example 5.3. For the permutation:

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 7 & 1 & 5 & 6 & 4 & 3 \end{pmatrix},$$

the global rule yields the same growth diagram as in 4.1. Thus we have the same bijection as before.

5.4 Local Rule on Edges

Finally we finish this section by introducing the fifth method for obtaining the bijection between S_n and a pair of standard tableaux of same shape $\lambda \vdash n$. A good reference for this method is Xavier Viennot's lecture videos (ch1b of [4]). The rule can be described by the following diagrams.

$$\begin{array}{c} 0 \\ \square \\ 0 \end{array} \quad \begin{array}{c} 1 \\ \square \\ 1 \end{array} \quad \begin{array}{c} j \\ \square \\ j \end{array} \quad \begin{array}{c} i+1 \\ \square \\ i \end{array}$$

In the third case assume $i \neq j$.

Example 5.4. Once again, consider the permutation:

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 7 & 1 & 5 & 6 & 4 & 3 \end{pmatrix}.$$

Then the local rules on edges yield us the following diagram figure 12.

Now we describe how to recover P and Q . The tableau Q is obtained inductively as follows. At step i , add a box labeled i to the end of the k_i -th row of the tableau constructed at the previous step, where k_i is the label on the i -th edge along the top boundary, read from left to right. Similarly, the tableau P is obtained inductively as follows. At step i , add a box labeled i to the end of the k_i -th row of the tableau constructed at the previous step, where k_i is the label on the i -th edge along the right boundary, read from bottom to top.

Once again we get the same pair of standard tableaux P and Q .

$$P = \begin{array}{|c|c|c|} \hline 1 & 3 & 6 \\ \hline 2 & 4 & \\ \hline 5 & & \\ \hline 7 & & \\ \hline \end{array}, \quad Q = \begin{array}{|c|c|c|} \hline 1 & 2 & 5 \\ \hline 3 & 4 & \\ \hline 6 & & \\ \hline 7 & & \\ \hline \end{array}.$$

Theorem 5.5. *All the methods defined so far—row bumping, the matrix-ball construction, the global rule, and the local rules on vertices and edges—yield the same bijection between elements of S_n and pairs of standard Young tableaux of the same shape $\lambda \vdash n$.*

Proof. One can refer [4] ch1b and ch1c for the equivalence of matrix ball construction (which Viennot call light and shadow method) and the local rule on edges and vertices. The equivalence of matrix ball construction and row bumping is done in chapter 4 of [2]. Now the equivalence of local rule on vertices and global rule on vertices is done in appendix A of [1]. □

	1	1	2	2	1	3	4
0	0	X 1	1	2	2	3	4
0	1	0	2	1	1	2	3
	0	0	0	0	X 1	1	1
0	1	0	2	1	0	2	3
	0	0	0	X 1	1	2	3
0	1	0	2	0	0	1	2
	0	0	0	0	0	X 0	2
0	1	0	2	0	0	0	1
	0	0	0	0	0	0	X 1
0	1	0	2	0	0	0	0
	X 1	1	2	2	2	2	2
0	0	0	1	0	0	0	0
	0	0	X 1	1	1	1	1
	0	0	0	0	0	0	0

Figure 12: Local rule on edges

6 Fomin's Growth Rules on the Young–Fibonacci Lattice

In this section, we study how the previously developed theory applies to the Young–Fibonacci lattice $\mathbb{Y}F$. Our goal is to use Fomin's local growth rule on vertices to construct a bijection between permutations in S_n and pairs of paths starting at 0 and ending at a common vertex on the n th level.

To construct the desired bijection, we first fix an R -correspondence on $\mathbb{Y}F$. By construction, any element $x \in \mathbb{Y}F$ at the n th level is covered by the elements $1x$ and $2z$, where z is an element at the $(n-2)$ th level. We define the R -correspondence by

$$\Phi_x(1x) = x, \quad \Phi_x(2z) = z.$$

Example 6.1. In this example we use the above R -correspondence and Fomin's local growth rule to obtain the pair of paths corresponding to the following permutation in S_9 . Let

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 5 & 4 & 9 & 1 & 8 & 6 & 7 & 3 \end{pmatrix}.$$

Applying the local rule on vertices, we obtain the following growth diagram (figure 13).

Thus the pair of paths corresponding to π is

$$P = (0, 1, 2, 12, 22, 221, 2211, 21211, 211211, 221211)$$

$$Q = (0, 1, 11, 21, 121, 221, 2211, 21211, 22211, 221211).$$

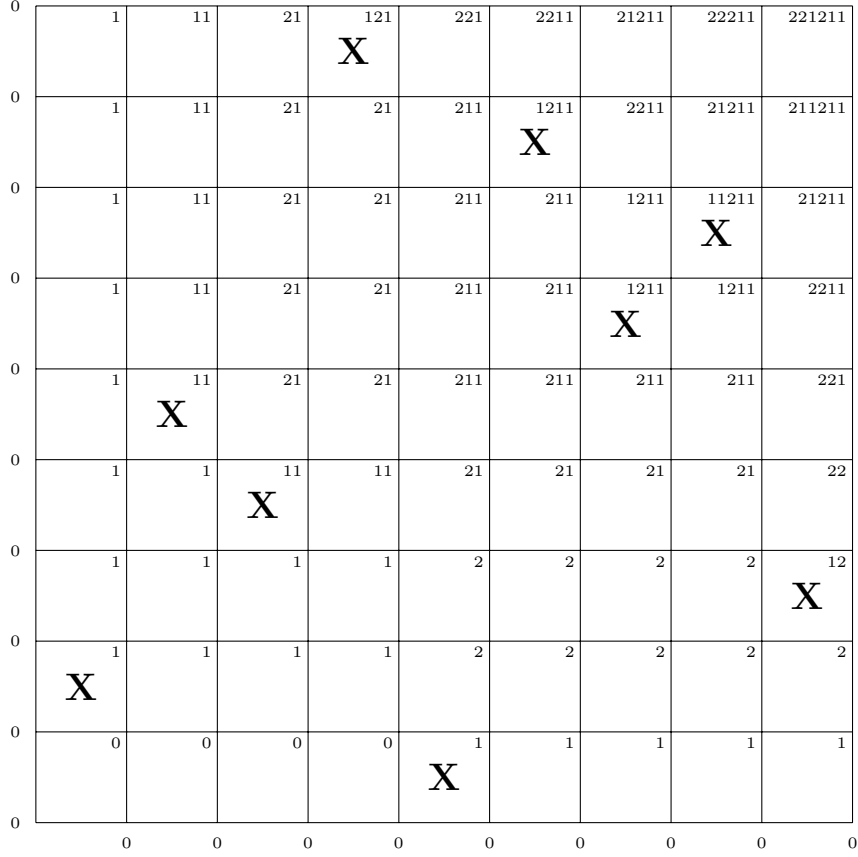


Figure 13: Growth diagram for the permutation π in the Young–Fibonacci lattice

7 Equivalent Constructions for the Young–Fibonacci Lattice

Analogous to the case of the Young lattice, here also we have several ways to obtain the bijection: a procedure similar to row bumping, along with a global growth rule on vertices. A key point is that these constructions—local rule, row bumping–type procedure, and global rule—are all equivalent and yield the same bijection. Most of the ideas will be illustrated through examples. Much of this section is more or less similar to the chapter 5 of [1].

Example 7.1. In the case of the Young lattice, every strictly increasing path from 0 can be represented by a standard tableau. It is natural to ask whether an analogous class of shapes, similar to Young diagrams, can be used to represent paths in this setting as well. One possible approach is to attempt the following construction.

We can represent an element of $\mathbb{Y}F$ typographically as a Young diagram as follows. Each “2” corresponds to two boxes stacked vertically, and each “1” corresponds to a single box; we proceed from left to right. With this convention, the shapes in Q above are:

$$Q = \left(0, \begin{array}{|c|} \hline \square \\ \hline \end{array}, \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array}, \begin{array}{|c|} \hline \square \\ \square \\ \hline \end{array}, \begin{array}{|c|} \hline \square \\ \square \\ \square \\ \hline \end{array}, \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array}, \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array}, \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \end{array}, \begin{array}{|c|c|c|c|c|} \hline \square & \square & \square & \square & \square \\ \hline \end{array}, \begin{array}{|c|c|c|c|c|c|} \hline \square & \square & \square & \square & \square & \square \\ \hline \end{array} \right)$$

It is natural to try to represent a pair of paths by:

$$P = \begin{array}{|c|c|} \hline 4 & 9 \\ \hline 3 & 8 \\ \hline \end{array} \begin{array}{|c|} \hline 2 \\ \hline \end{array} \begin{array}{|c|c|c|} \hline & & \\ \hline 7 & 1 & 6 \\ \hline \end{array} \begin{array}{|c|} \hline 5 \\ \hline \end{array} \quad Q = \begin{array}{|c|c|} \hline 5 & 8 \\ \hline 4 & 7 \\ \hline \end{array} \begin{array}{|c|} \hline 3 \\ \hline \end{array} \begin{array}{|c|c|c|} \hline & & \\ \hline 9 & 2 & 6 \\ \hline \end{array} \begin{array}{|c|} \hline 1 \\ \hline \end{array}$$

These kind of shapes (shapes corresponding to paths in $\mathbb{Y}F$) are called *path tableaux*. One can do this, but there are several disadvantages compared with the tableaux for Young’s lattice.

While each entry in the top row must be greater than the one below it, there are no further obvious rules governing admissible numberings for a fixed shape. In general, a numbering satisfying this condition need not correspond to a path in $\mathbb{Y}F$. For example, the tableau

$$T = \begin{array}{|c|c|c|c|} \hline 6 & 2 & & 7 \\ \hline 5 & 1 & 3 & 4 \\ \hline \end{array}$$

does not represent a path, since 2212 does not cover 2211 in $\mathbb{Y}F$. Another method for representing paths in $\mathbb{Y}F$ will be described in the next section.

7.1 Insertion Algorithm

Similar to the insertion algorithm in Young lattice, we have an algorithm for $\mathbb{Y}F$ as well. The algorithm described below produces the same output as Fomin's approach. Suppose we want to insert a number a into a "partial tableau" T of shape x . We start by comparing a with the value t_1 at the leftmost cell of the bottom row of T . If $a > t_1$, we attach a new cell to the left of t_1 , place a in it, and stop; the resulting tableau has shape $1x$. If $a < t_1$, we move a into the cell immediately above t_1 , displacing any element b that may already occupy that cell. If that cell was empty, the process terminates. Otherwise, we repeat the procedure inductively: b is now compared with the element t_2 lying immediately to the right of t_1 in the bottom row. It is not difficult to show that the insertion algorithm respects the covering relations.

Example 7.2. Inserting 8 into the partial tableau

$$\begin{array}{|c|c|c|} \hline 1 & 4 & \\ \hline 9 & 5 & 2 \\ \hline \end{array} \quad \text{we get} \quad \begin{array}{|c|c|c|c|} \hline 8 & 1 & & \\ \hline 9 & 5 & 4 & 2 \\ \hline \end{array}$$

Example 7.3. Consider

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 5 & 4 & 9 & 1 & 8 & 6 & 7 & 3 \end{pmatrix}.$$

Then starting from the empty tableau and inserting the integers along the second row of the permutation, we get the following tableau.

$$\begin{array}{|c|} \hline 2 \\ \hline \end{array}, \quad \begin{array}{|c|c|} \hline 5 & 2 \\ \hline \end{array}, \quad \begin{array}{|c|c|} \hline 4 & \\ \hline 5 & 2 \\ \hline \end{array}, \quad \begin{array}{|c|c|c|} \hline 4 & & \\ \hline 9 & 5 & 2 \\ \hline \end{array}, \quad \begin{array}{|c|c|c|} \hline 1 & 4 & \\ \hline 9 & 5 & 2 \\ \hline \end{array}, \quad \begin{array}{|c|c|c|c|} \hline 8 & 1 & & \\ \hline 9 & 5 & 4 & 2 \\ \hline \end{array},$$

$$\begin{array}{|c|c|c|c|c|} \hline 6 & & 1 & & \\ \hline 9 & 8 & 5 & 4 & 2 \\ \hline \end{array}, \quad \begin{array}{|c|c|c|c|c|} \hline 7 & 6 & 1 & & \\ \hline 9 & 8 & 5 & 4 & 2 \\ \hline \end{array}, \quad \begin{array}{|c|c|c|c|c|c|} \hline 3 & 7 & & 1 & & \\ \hline 9 & 8 & 6 & 5 & 4 & 2 \\ \hline \end{array}.$$

Our final tableau P (represents the path from before) and its recording tableau Q looks like

$$P = \begin{array}{|c|c|c|c|c|c|} \hline 3 & 7 & & 1 & & \\ \hline 9 & 8 & 6 & 5 & 4 & 2 \\ \hline \end{array} \quad Q = \begin{array}{|c|c|c|c|c|c|} \hline 5 & 8 & & 3 & & \\ \hline 4 & 7 & 9 & 2 & 6 & 1 \\ \hline \end{array}$$

Note that the recording tableau Q is same as the path tableau Q we obtained in example 7.1.

Now there is no obvious way to retrieve the path from the tableau P . The next definition of tableaux is motivated from P in the above example.

Definition 7.4. The shapes with fillings such that elements decrease along the bottom row and up each column are called *tableaux*.

There are no requirements on the top row elements. The advantage of tableaux over path tableaux is that we can easily write down all legitimate tableaux of a given shape. Let $ZT(x)$ (resp. $PZT(x)$) denote the set of all tableaux (resp. path tableaux) of shape x . Set $|z_x| := |ZT(x)|$.

Proposition 7.5. *There is a bijection between $\pi \in S_n$ and pairs (P, Q) with $P \in ZT(x), Q \in PZT(x)$ and $|x| = n$. We write it as $\pi \xleftrightarrow{F-R-S} (P, Q)$.*

It is easy to see that the insertion algorithm yields a legitimate tableau P and the recording tableau $Q \in PZT(x)$. Thinking the process to be inductive, we can show that insertion algorithm is invertible. For a rigorous proof, refer proposition 5.2.4 of [1].

The following lemma asserts that the bijection we have via Fomin's method or insertion algorithm are the same. The proof of this is slightly technical and is done in lemma 5.4.4 of [1].

Lemma 7.6. *Let $\pi \xleftrightarrow{F-R-S} (P, Q)$. Then (P, Q) represents the same pair of paths $(\hat{P}, \hat{Q})$ obtained by Fomin's method.*

As an immediate consequence of the lemma 7.6, we have the following theorem (same proof we did for Young lattice).

Theorem 7.7. *If $\pi \xleftrightarrow{F-R-S} (P, Q)$, then $\pi^{-1} \xleftrightarrow{F-R-S} (Q, P)$.*

Now let us understand the theorem. It is natural to ask what this theorem mean? i.e., if π corresponds to $(P, Q), P \in ZT(x), Q \in PZT(x)$ how does (Q, P) represents the inverse of π , because Q might not be in $ZT(x)$. The theorem is clear when P and Q are considered as paths. The theorem states that any path in $\mathbb{Y}F$ from 0 to x in can be written in either of the form; as an element of $PZT(x)$ or $ZT(x)$. For example if we have $P \in ZT(x)$, then to view $P \in PZT(x)$, we can choose any $Q \in PZT(x)$ and work backwards to get a permutation π and then apply the algorithm to π^{-1} to get the pair (A, B) . Then $B \in PZT(x)$ represents P in the required form.

Corollary 7.8. $\sum_{|x|=n} z_x^2 = n!$ and $\sum_{|x|=n} z_x = \#\{\text{involutions in } S_n\}$.

Example 7.9. Consider the involution

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 2 & 1 & 7 & 6 & 5 & 4 \end{pmatrix}.$$

Then applying insertion algorithm we obtain the tableau:

$$P = \begin{array}{|c|c|c|} \hline 4 & 5 & 1 \\ \hline 7 & 6 & 3 \\ \hline \end{array} \begin{array}{|c|} \hline 2 \\ \hline \end{array}$$

Observe that the cycle decomposition for π is

$$\pi = (13)(2)(47)(56)$$

which corresponds exactly to our P . Is this a coincidence? Interestingly, No! We have the following theorem.

Theorem 7.10. *Let $\pi \in S_n$ be an involution and $\pi \xleftrightarrow{F-R-S} P \in ZT(x)$. Then P represents the cycle decomposition of π in an obvious manner.*

We shall do the proof of this theorem in next section.

7.2 Global Rule for Young Fibonacci Lattice

Let π be a generalized permutation of the cells of Sq_n . Then we define a function G which associates each vertex v of Sq_n an element $G(v) \in \mathbb{Y}F$ recursively as follows. Start with the empty word 0. Fix a vertex v . Now let R (resp. C) be the cell of π which is left and below to v in the highest occurring row (resp. column). If $R = C$, append a 1 to the end of w and if $R \neq C$, then append a 2. Now delete R and C from π and repeat the process. When all the cells of π lying left and below to v is exhausted, the process terminates and the end word is $G(v)$.

Example 7.11. Consider the permutation

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}.$$

Then the global algorithm produces the following diagram:

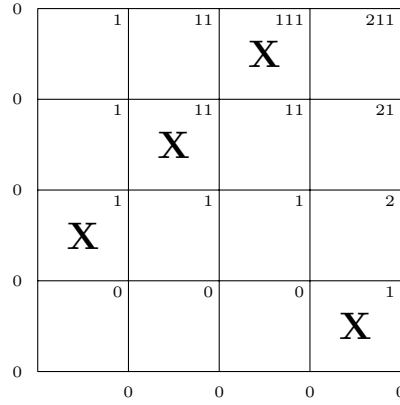


Figure 14

It is worthy to note that the global algorithm yields the same diagram as that of Fomin's local growth rules. The interested reader can try the example 6.1 to see that the local rules and global rules agree for the permutation π (figure 13).

The following theorem asserts the equivalence of global and local rule. Thus we have the equivalence of three methods to obtain the bijection between S_n and a pair of paths starting from 0 and ending at the same vertex at the n th floor. For a proof of the following theorem, refer lemma 5.4.3 of [1].

Theorem 7.12. *Let π be a generalized permutation of the cells of Sq_n , and let $g : Sq_n \rightarrow \mathbb{Y}F$ be the growth corresponding to π with the R -correspondence defined in section 6. Then we have $g(v) = G(v)$ for every vertex v of Sq_n .*

Now we do the proof of the theorem 7.10.

proof of theorem 7.10 . We use the global description $G(v)$ at each v along the upwards diagonal of Sq_n . Because π is symmetric about this diagonal (since an involution), each cell in π pair up with its image and fixed points of π will contribute to the columns of height 1. $\square$

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